Expert AI Panels for Creative Work Evaluation: Evidence from Puzzle Hunt Design

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Abstract

Evaluating creative work requires the kind of domain expertise that is scarce, expensive, and difficult to apply consistently at scale. This paper reports an unexpected finding about structured evaluation: a rubric applied without expert identity context systematically *inverts* the quality hierarchy, producing higher scores for accessible work than for ambitious work. Across a fifteen-puzzle corpus spanning three quality tiers, rubric-only evaluation yields a pass rate of 80% for Standard Hunt puzzles but only 40% for MIT/Elite-tier puzzles—because elite puzzles deliberately sacrifice Clarity and Solvability for Elegance and Reading Reward, and an equal-weight rubric treats that trade-off as a defect. Expert identity, encoded in rich reviewer profiles, reintroduces the calibration that converts raw dimension scores into verdicts appropriate to the paradigm under review.

We ground this investigation in puzzle hunt design, a collaborative creative practice with well-defined quality criteria, a recognized expert community, and the property that answer correctness is binary while quality is contested. We assemble panels of AI reviewers instantiated from persistent, domain-specific profiles—structured markdown documents capturing each simulated reviewer's design philosophy, evaluative lens, and named evaluative frameworks—and compare eight conditions in a controlled ablation across 15 real published puzzles from MIT Mystery Hunt 2023, Huntinality, and Teammate Hunt.

The eight-condition ablation reveals two additional findings. First, a *feedback quality gradient*: richer profiles surface six analytical frameworks (*a-ha economy*, *load-bearing test*, *world-model consistency*, *social fabric*, *perceptual-shift*, *visual grammar*) that generic profiles and bare prompting do not introduce; these frameworks appear in C6 reviews but in no input text provided to the reviewers. Second, adding the *Riven Standard* as a scored seventh dimension ("the puzzle IS what the field does, not an obstacle overlaid on it") achieves a 24.5 percentage-point tier-separation gap between elite and magazine-tier puzzles, compared to 10.9 points for the equal-weight baseline.

All results are characterized as valid within the authors' constructed evaluation framework; a human calibration study remains future work.

We release the full profile library, scoring rubric, and replication corpus to support adaptation to other creative evaluation domains.

1 Introduction

A puzzle hunt presents its participants with a peculiar evaluative duality: the answer to any given puzzle is either correct or it is not, but the puzzle itself is either elegant or it is not, and that second judgment—the one that separates a puzzle worth solving from one that merely has a solution—is far harder to make and far harder to teach.

1.1 The Problem: Domain Expertise Is Scarce

Evaluating creative work well requires more than intelligence and goodwill. It requires the accumulated vocabulary of a practice: the ability to name what is wrong, to point to the specific mechanism that fails, to distinguish a problem of clarity from a problem of elegance from a problem of fairness. In creative domains with recognized expert communities, this vocabulary is hard-won—it takes years of engagement with a tradition to develop the evaluative lens that allows a practitioner to say not just "this puzzle is unsatisfying" but "the confirmation mechanic is ambiguous, which shifts the cognitive load from *aha* to *relief*, and that is a fairness failure, not a difficulty setting." Such expertise is scarce, expensive to access, inconsistently applied, and largely unavailable at the scale required for iterative creative pipelines. The consequence is that early-stage creative work—when the most is still malleable—is typically evaluated by creators themselves, by generalist colleagues, or not at all. Domain expert review arrives late, if at all, and is difficult to incorporate systematically.

1.2 The Insight: Persona Depth Is the Variable That Matters

Large language models have demonstrated capacity to simulate domain knowledge across a wide range of fields [?], and recent work on AI-simulated review has shown that LLMs can approximate expert academic judgment under controlled conditions [? ?]. However, the standard approach to persona simulation—supplying a brief role description such as "you are an experienced puzzle designer"—produces correspondingly generic output. A bare prompt elicits the evaluative vocabulary the model associates with the role label, not the specific, idiosyncratic frameworks that distinguish one expert's perspective from another's. The result is feedback that is superficially domain-appropriate but operationally thin: it identifies that problems exist without naming the mechanisms that produce them, and it generates recommendations too vague to act on.

The insight driving this work is that the quality of AI-simulated expert feedback is not a fixed property of the underlying model—it is a function of the richness of the persona description. When a reviewer profile captures not just a role label and a list of interests but a full evaluative lens, including characteristic frameworks, named concerns, design philosophy, and the vocabulary the reviewer reaches for when a puzzle misfires, the resulting simulated feedback is qualitatively different: more specific, more transferable, and more actionable. The profile is not window-dressing—it is the mechanism by which the model instantiates a distinctive evaluative stance rather than averaging across all evaluative stances it has encountered in training.

1.3 Our Approach: Persistent Profiles and Structured Evaluation

We operationalize this insight through a system of *persistent expert profiles*: structured markdown documents of 80–120 lines each that capture a simulated reviewer's design philosophy, evaluative lens, characteristic concerns, analytical vocabulary, and voice. Profiles are loaded once per review session and reused across puzzles, building on the token-efficient profile architecture developed in prior work [**?**]. A panel of three reviewers is assembled from the profile library for each puzzle under evaluation, and each reviewer scores the puzzle on a six-dimension rubric (Clarity, Solvability, Elegance, Reading Reward, Fun, Confirmation) with a 1–5 scale per dimension, yielding a total out of 30 with a passing threshold of 22. Scores are accompanied by written feedback and a set of specific, actionable fix recommendations.

We evaluate this system empirically against two lower-profile conditions: a bare-prompt baseline with no reviewer persona, and a generic profile set drawn from an earlier 12-profile library [**?**]. The comparison isolates the effect of profile depth from the effect of having a profile at all, allowing us to attribute outcome differences specifically to domain specificity rather than to the presence of any persona description. The testbed is the Age of Empires puzzle hunt, a set of original puzzles designed around the thematic and mechanical universe of the Age of Empires game franchise. All three puzzles evaluated in the main experiment were developed independently of the reviewer profiles, ensuring that the profiles do not encode any information about the specific artifacts being evaluated.

1.4 What Is a Puzzle Hunt?

For readers unfamiliar with the domain: a *puzzle hunt* is a competitive or collaborative event in which teams solve a sequence of puzzles, each yielding an answer—typically a word or phrase—that feeds into a larger *meta-puzzle* resolving the hunt's central mystery. The puzzles themselves range across a wide variety of types (word puzzles, logic grids, cryptic crosswords, visual puzzles, knowledge challenges) and are evaluated against a demanding set of craft criteria: they must be unambiguously solvable by the target audience, must reward careful reading, must produce an *a-ha* moment at the point of solution, and must feel elegant and fair in retrospect. The hunt community has developed a sophisticated evaluative vocabulary around these criteria, and recognized practitioners—editors at MIT Mystery Hunt, MITMH-adjacent events, and independently organized hunts—bring a distinctive and largely codified expert perspective to the work [**?**]. This makes puzzle hunt design an unusually appropriate testbed for studying AI-simulated expert review: the expertise is real, specialized, and reasonably well-documented, but access to it is constrained by the small size and geographic concentration of the expert community.

1.5 Contributions

This paper makes the following contributions:

1. **The calibration inversion.** Rubric-only evaluation (no reviewer identity) inverts the quality tier hierarchy: across 15 real published puzzles, it produces an 80% pass rate for Standard Hunt puzzles but only 40% for MIT/Elite-tier puzzles. The mechanism is traceable: elite puzzles deliberately sacrifice Clarity and Solvability for Elegance and Reading Reward, and an equal-weight rubric treats that trade-off as a defect. This finding is a concrete warning for practitioners—rubric-only evaluation appears objective but is miscalibrated for ambitious creative work.

2. **Profile-based AI expert panel system.** A complete system for assembling and deploying AI reviewer panels for creative work evaluation, built on persistent structured profiles encoding domain-specific evaluative frameworks. The system is described in sufficient detail to support replication and adaptation to other creative domains. *Note: all quantitative findings are valid within the authors' constructed evaluation framework; the rubric, tier assignments, and thresholds were constructed by the research team and have not been externally calibrated against a held-out human expert standard.*

3. **Empirical eight-condition profile-depth ablation.** A controlled experiment across 15 real published puzzles from MIT Mystery Hunt 2023, Huntinality, and Teammate Hunt, comparing conditions from bare prompting to full domain-specific profiles to profile plus principles. The ablation isolates the contribution of domain specificity, profile component type, and rubric design.

4. **Taxonomy of surfaced analytical frameworks.** Six named evaluative frameworks (*world-model consistency*, *a-ha economy*, *load-bearing test*, *social fabric*, *perceptual-shift*, *visual grammar*) that appear in C6 reviews but in no input text provided to the reviewers, constituting a contribution to the vocabulary of puzzle design evaluation.

5. **Open profile library and replication package.** The full set of 29 domain-specific reviewer profiles (C8 format, $\sim$75 lines), the seven-dimension

weighted rubric, all review transcripts, and analysis scripts, released to support replication and adaptation to other creative evaluation domains.

1.6 Paper Organization

Section **??** situates this work relative to AI-simulated review, persona-based prompting, computational creativity evaluation, and the emerging literature on puzzle design as a research domain. Section **??** describes the profile architecture, scoring rubric, and experimental design in detail. Section 4 reports results from the profile ablation and cross-hunt generalizability experiments. Section 5 interprets the results and discusses limitations. Section 6 summarizes contributions and directions for future work.

2 Related Work

2.1 AI-Simulated Expert Review

The use of LLMs to simulate peer review has attracted growing attention as a mechanism for scaling evaluation capacity. Shah and Wein [**?**] investigate whether LLMs can replicate human reviewer judgments on academic manuscripts, finding moderate agreement on surface-level criteria (clarity, presentation) but weaker alignment on deeper assessments of novelty and significance. Liu et al. [**?**] present ReviewerBot, a system that generates structured academic reviews by conditioning on paper content and area-specific rubrics, demonstrating that domain-targeted prompting improves the specificity of feedback relative to generic review generation. The broader "LLM-as-judge" paradigm [**?**] has shown that LLM evaluators can achieve reasonable agreement with human panels on pairwise preference tasks, though with known biases toward verbosity and self-generated content.

Most directly relevant to the present work is **?**], which introduces token-efficient persistent reviewer profiles for academic paper evaluation. That work treats profiles as reusable artifacts with structured schemas, demonstrating that explicit persona documentation reduces per-call token overhead while preserving reviewer consistency across sessions. However, the academic review setting shares a critical property: evaluation criteria are relatively standardized. Conference rubrics specify what "novelty," "soundness," and "presentation" mean, and the broader community has converged on shared definitions over decades of practice. This standardization makes academic review a tractable first testbed but limits how far findings generalize.

Creative domain evaluation presents a structurally different challenge. There is no shared rubric for what makes a puzzle elegant, a level design satisfying, or a narrative resonant. Quality is operationalized through expert judgment, and expert judgment is itself the object of inquiry. Domain knowledge functions not as background context for applying a known rubric, but as the mechanism through which evaluation criteria are derived in the first place. We extend the AI-simulated review paradigm to this harder setting, showing that profile specificity — not merely profile presence — is the lever that matters when criteria cannot be assumed.

2.2 Persona-Based Prompting

Persona-based prompting has a substantial prior literature across several application areas. Zhang et al. [?] demonstrate that injecting persona descriptions into dialogue systems improves response coherence and user engagement, establishing that LLMs benefit from explicit identity framing rather than generic instruction. Park et al. [?] extend this to generative agents with persistent identity representations, showing that memory-augmented personas can sustain coherent behavior across multi-turn interactions in simulated social environments. Dang et al. [?] apply persona conditioning to creative writing assistance, finding that distinct persona instantiations produce meaningfully different stylistic outputs from the same model, with implications for diversity in co-creative systems.

Across this literature, persona is treated primarily as a *binary* variable: either a persona description is present or it is not. The implicit assumption is that any reasonable persona specification will activate the relevant prior knowledge in the underlying model. What remains unstudied is persona quality as an independent variable — whether the richness, specificity, and internal coherence of a persona description materially affects the quality of the output it produces, and whether this effect can be measured and replicated.

This gap is the central question of the present work. We treat persona profiles as design artifacts with measurable quality levels, comparing a baseline condition (no profile), a generic expert condition (v1: 12 profiles, ~50 lines each), and a domain-specific condition (v2: 29 profiles, ~80–120 lines each, with named analytical frameworks). This design allows us to decompose the contribution of persona presence from the contribution of persona quality, and to characterize the dimension along which quality improvements manifest — not primarily in score magnitude, but in the specificity and transferability of the evaluative frameworks introduced.

2.3 Creative Evaluation Methodology

Evaluating creative artifacts is a foundational problem in computational creativity research. Colton and Wiggins [?] survey the field's evaluation challenges, arguing that creativity assessment requires distinguishing novelty, quality, and intentionality — dimensions that resist reduction to scalar metrics and that depend on the evaluator's cultural and domain context. Ritchie [?] proposes a formal criterion-based framework for assessing computational creativity systems, but notes that any such framework must be instantiated with domain-specific quality standards rather than universal ones. Boden [?] distinguishes combinational, exploratory, and transformational creativity, with the latter category in particular resisting automated evaluation because the evaluator must recognize when a new conceptual space has been opened.

Within game design, quality criteria are contested but not unarticulculated. Sicart [?] argues that proceduralist design frameworks — rubrics centered on rule systems and optimal play — miss the ethical and aesthetic dimensions of gameplay experience. Juul [?] develops a theory of games as half-real, emphasizing the interaction between formal rule structures and player-projected fiction. These perspectives suggest that game design evaluation requires sensitivity to both structural properties (is the mechanism sound?) and experiential ones (does it produce the intended affect?).

Puzzle design in particular has a practitioner tradition of quality criteria that is more explicit than most creative domains. The Riven Standard, articulated by Rand Miller in reflecting on the *Myst* sequel, holds that a well-designed puzzle is one that *is* what the domain does: its mechanic should be inseparable from its setting, not grafted onto it [?]. Snyder [?] enumerates craftsmanship criteria for crossword construction that translate across puzzle types: every element should pull its weight, solutions should be uniquely determined, and the solver's "aha" should be earned rather than forced. Our six-dimension scoring rubric (Clarity, Solvability, Elegance, Reading Reward, Fun, Confirmation) operationalizes these practitioner principles into measurable dimensions while preserving their emphasis on the solver's experiential arc. The rubric's contribution is not to replace qualitative judgment but to give AI reviewer panels a stable coordinate system within which qualitative observations can be located and compared.

2.4 Puzzle Hunt as a Research Domain

Puzzle hunts occupy an unusual position in the landscape of creative artifacts: they are both genuinely creative and partially verifiable. Each puzzle has a single correct answer, so correctness can be confirmed objectively. But whether a puzzle is *good* — fair, elegant, thematically resonant, and appropriately difficult — is a matter of expert judgment that the puzzle hunt community has developed sophisticated vocabulary for debating. This combination of verifiable correctness and contested quality makes puzzle hunt design a particularly tractable testbed for empirical work on creative evaluation.

Several properties distinguish puzzle hunts from superficially similar domains. Unlike trivia, where domain knowledge determines whether a solver can answer a question, puzzle hunt design requires that domain knowledge *be the mechanism*: the puzzle should be impossible without understanding the domain, but fully solvable once one does. This structural property — sometimes called thematic integration — is itself a quality dimension that skilled evaluators assess and that naive evaluators tend to miss [?]. Unlike narrative fiction or visual art, hunts have a documented expert community (Mystery Hunt constructors, BAPHL designers, MITMH veterans) whose evaluative philosophies are discussed publicly in post-hunt retrospectives, making it possible to construct profiles grounded in real critical perspectives rather than invented ones.

The empirical study of puzzle design as creative practice is small but growing

[? ?]. Prior work has examined solver behavior, difficulty calibration, and hint design, but has not examined AI-assisted evaluation or the role of reviewer persona in shaping evaluation quality. Our contribution is to establish puzzle hunt design as a controlled creative evaluation testbed: one with sufficient domain structure to support empirical comparison, sufficient creative latitude to make evaluation non-trivial, and sufficient community documentation to ground AI reviewer profiles in authentic expert perspectives.

3 Methodology

3.1 System Architecture

Our system instantiates a small panel of AI-simulated domain experts to evaluate puzzle hunt puzzles using a structured scoring rubric. The pipeline proceeds in five stages: (1) a *profile library* of persistent expert reviewer profiles is maintained as version-controlled markdown artifacts; (2) for each puzzle under evaluation, a *panel assembly* step selects three reviewers from the library with complementary evaluative lenses (e.g., structural rigor paired with accessibility and narrative); (3) each selected profile is loaded as full system context and the model is prompted to generate a written *review* from that expert's perspective; (4) the three reviews are *synthesized* to surface consensus findings, unresolved disagreements, and actionable revisions; (5) a *score* on six or seven dimensions is extracted (see Section 3.3), producing a per-puzzle quality profile (see Figure 1 for the full pipeline diagram). All experiments reported here use Claude Sonnet 4.6 as the underlying model, with temperature 1.0 and no output truncation. Profiles are persistent artifacts — they are authored once, refined over time, and loaded unchanged for each review session. No profile content is regenerated or conditioned on the puzzle being evaluated.

3.2 Profile Design

Each reviewer profile is a markdown document of 80–120 lines organized into six sections: **Identity** (name, affiliation, and credentials as a puzzle or game designer); **Design Philosophy** (3–4 paragraphs articulating the reviewer's core convictions in first-person or closely attributed voice); **Review Lens** (a structured list of 10–14 specific evaluative questions the reviewer applies to every puzzle); **Key Works** (titles and roles that ground the persona in documented output); **Puzzle Hunt Credentials** (competitive and constructive history relevant to the hunt domain); and **Key Sources** (links and publications anchoring the profile to verifiable public record).

The *review lens* section is the profile's primary functional contribution: it operationalizes the reviewer's philosophy as a checklist of domain-specific probes that the model executes during evaluation. Table 1 summarizes the structural differences between the two profile versions used in our experiments.

The key design decision distinguishing v1 from v2 is the introduction of

Table 1: Profile version comparison. Named frameworks are evaluative concepts introduced in the review lens with specific names (e.g., *load-bearing test*, *a-ha economy*, *world-model consistency*).

Version	Profiles	Avg. lines	Named frameworks	Characteristic concerns
v1	12	∼50	No	No
v2	29	∼95	Yes	Yes

named evaluative frameworks — domain-specific analytical concepts that the reviewer applies by name, enabling the model to produce critique that is transferable and citable rather than generic. To make this concrete, consider the same reviewer (Thomas Snyder) across both versions. In v1, the analogous profile entry reads: "Checks that each element is necessary and that the puzzle has a clear intended solve path." In v2, the corresponding review lens asks: "Is each element load-bearing? He strips individual constraints, given values, or visual elements and tests whether the puzzle still forces a unique solution. Anything that can be removed without consequence was not design — it was accumulation." The v2 formulation names a test (*load-bearing test*), specifies the operation (remove and check uniqueness), and characterizes failure in terms the reviewer would use in actual feedback. Similar named frameworks introduced in v2 include the *a-ha economy* (does the puzzle produce one satisfying moment of insight, or squander it through redundancy), *world-model consistency* (does the puzzle's logic hold within the fictional or factual universe it invokes), *social fabric* (does the puzzle work in a team-solving context or require solitary reconstruction), and *perceptual-shift* (does the solver experience a genuine reframing of the problem, or merely accumulate information). The presence of these frameworks in review output, and their absence in v1 output, is the primary signal we measure in Experiment 0.

3.3 Scoring Rubric

Puzzles are scored on seven dimensions by each of the three panel reviewers. The rubric design is itself an empirical contribution of this paper: we begin with a six-dimension equal-weight baseline (used in Experiments 0–2 to control for rubric effects while testing profile depth), then validate a seven-dimension weighted configuration in Section 4.4. Table 2 provides full definitions for all seven dimensions.

Clarity (×1): whether a solver can understand what the puzzle is asking without external help. Hunt puzzles intentionally withhold genre labels; the mechanism must do all signposting.

Solvability (×1): whether a solver with the target domain knowledge can reliably reach the correct answer. Distinct from difficulty: a puzzle can be challenging and fully solvable, or easy and broken.

Elegance (×2 in validated rubric): whether the puzzle's mechanism feels clean, satisfying, and non-arbitrary. Elegance signals intentional design; we distinguish elegant puzzles from merely correct ones. Double-weighted in the validated rubric because it is empirically the strongest tier discriminator across 15 puzzle types.

Reading Reward (×2 in validated rubric): whether domain knowledge is load-bearing. A puzzle solvable by general reasoning without domain knowledge has failed its primary design criterion. Double-weighted alongside Elegance; both are proxies for the Riven Standard (see below).

Fun (×1): whether the experience is enjoyable and produces an identifiable *aha* moment. The primary phenomenological output of puzzle design.

Confirmation (×1): whether the solver knows when they have reached the correct answer without being told. Weak confirmation produces anxiety and hedged submissions.

Riven Standard (×1, validated rubric only): the field's highest-order quality criterion, named after Rand Miller's design principle [**?**]: "The puzzle IS what the field does, not an obstacle overlaid on it." Scored on a five-point scale from 1 (*theme is decorative; any content could substitute*) to 5 (*mechanic is inseparable from the domain and could not exist in any other context*). Elegance and Reading Reward are proxies for this criterion; adding it explicitly captures residual variance not explained by the two weighted proxies. Riven Standard scores reported in Section 4.4 were inferred by the research team from existing C6 review texts rather than produced through independent blind scoring; a formal inter-rater reliability study is reserved for future work.

The six-dimension equal-weight version (/30, pass ≥ 22) is used in Experiments 0–2 as the controlled baseline rubric. The seven-dimension weighted version (/45, pass ≥ 33, with Elegance and Reading Reward each counted twice) is validated in Section 4.4 against 15 real puzzle hunt puzzles spanning three quality tiers; it achieves 24.5 percentage-point tier separation versus 10.9pp for the equal-weight baseline. Both thresholds correspond to approximately 73% of the maximum, calibrated to the quality level at which puzzles in well-regarded MIT Mystery Hunt and Microsoft Puzzlehunt rounds have shipped. This threshold was calibrated against the authors' prior experience with puzzle hunt quality and has not been validated against a held-out set of human expert ratings. All pass/fail verdicts reported in this paper should be interpreted as valid within this author-constructed framework.

The three-reviewer design provides lens diversity (structural rigor, narrative/accessibility, domain specificity) while remaining computationally tractable. Because all reported scores represent single evaluation runs at temperature 1.0, reported scores should be understood as point estimates without sampling variance. The score ranges produced by the three-reviewer panel provide a partial proxy for evaluative uncertainty; we report these ranges when spread exceeds two points on any dimension. High variance signals genuine evaluative disagreement that averaging would suppress.

3.4 Testbed: Puzzle Hunt Design

A *puzzle hunt* is a collection of puzzles, each with a unique verifiable answer (typically a word or short phrase), organized so that earlier answers unlock later puzzles and feeder answers combine to solve a final *meta-puzzle*. Hunt puzzles are distinguished from crosswords or trivia by their use of domain knowledge as a *load-bearing mechanism*: the puzzle's core step requires understanding a specific field — a game, a musical corpus, a fictional universe, a historical event — and cannot be completed by general reasoning alone. The solver must know the domain to solve the puzzle; the puzzle structure verifies that they do.

This makes puzzle hunt design an unusually tractable testbed for creative evaluation research, for three reasons. First, answers are objectively verifiable: there is a single correct extraction, and whether it produces the intended answer word is a binary check. This removes the most common objection to AI creative evaluation — the absence of ground truth. Second, the expert practitioner community is documented and its evaluative language is on record: published interviews, design postmortems, and instructional texts by constructors such as Thomas Snyder, Mike Selinker, Rand Miller, Wei-Hwa Huang, Patrick Berry, and Dan Katz provide ground truth for the evaluative frameworks our profiles encode. Third, the domain admits natural variation across hunt types, allowing cross-domain generalizability studies without leaving the testbed.

We use three hunt scenarios as our experimental corpus. **Age of Empires** ("Wololo") is a game-knowledge domain hunt built around the *Age of Empires* real-time strategy series, with 5 feeder puzzles and a meta; puzzles require knowledge of civilizations, units, buildings, and game mechanics from the series. **Wavelength** is a music domain hunt with 6 puzzles requiring identification of songs, artists, albums, and musical relationships. **Dead Reckoning** is a science-fiction instrument-panel domain hunt with a projected corpus of 15 puzzles, set in a fictional starship universe with canonical reference data maintained in a locked world file; 3 puzzles were available at the time of writing with additional puzzles in progress. Puzzles, not hunts, are the unit of analysis throughout our experiments: each puzzle is evaluated independently by its three-reviewer panel, and aggregate hunt-level scores are computed from puzzle-level results.

3.5 Experimental Design

We report results from three experiments, one of which has complete data and two of which are in progress at the time of submission.

Experiment 0 (Profile Ablation) is the primary empirical contribution of this paper. We select three puzzles from the Age of Empires corpus (AoE-I, AoE-II, AoE-III, corresponding to the Dark Age, Feudal Age, and Castle Age rounds) and evaluate each under three conditions. *Condition 0* is a bare prompt: the model receives the puzzle text and the instruction "You are an expert reviewer. Score this puzzle on the following 6 dimensions." No persona context is provided. *Condition 1* is the v1 profile panel: three reviewers are selected from the 12 v1 profiles and their abbreviated persona descriptions are loaded as system context. *Condition 2* is the v2 profile panel: three reviewers are selected from the 29 v2 profiles with full 80–120-line profiles as system context. The three conditions share identical puzzle inputs, identical scoring rubrics, and identical synthesis prompts; only the profile context varies. We measure four outcomes: per-dimension score, feedback word count, named analytical frameworks introduced per review (zero in Condition 0 by construction; counted automatically from review text in Conditions 1 and 2), and actionable revision suggestions per review (manually coded into specific/general/absent). Figure 2 will present score by condition across all six dimensions; Table 3 will report full per-puzzle, per-reviewer breakdowns.

Experiment 1 (Profile Upgrade) is the Condition 1-to-Condition 2 arm of Experiment 0, for which complete data is already available. We report these results separately in Section 4 to allow direct comparison with the published v1 evaluation baseline, noting that Experiment 1 results are a strict subset of the Experiment 0 design. The key finding — a +0.57 average score delta from v1 to v2 across the three puzzles, with materially larger gains in feedback specificity — is discussed in detail in Section 4.1.

Experiment 2 (Cross-Hunt Generalizability) extends the v2 panel evaluation to 25 puzzles across all three hunt types: the five Age of Empires puzzles (AoE-I through AoE-V), six Wavelength puzzles, and a subset of Dead Reckoning puzzles available at evaluation time. All puzzles are evaluated under Condition 2 only. The primary questions are whether the evaluation quality gain observed in the Age of Empires corpus generalizes to music and science-fiction domain puzzles, and which named analytical frameworks transfer across domain types. Results will be reported as score deltas relative to each puzzle's v1 baseline where available, and as raw v2 scores for puzzles with no prior evaluation. Figure 4 will present score by hunt type; Table 4 will summarize hunt-level results. Data collection is ongoing; results presented in Section 4.2 reflect available data at submission time.

Experiment 3 (Same-Input Divergence) addresses a different question: given identical world-building inputs, does the pipeline produce structurally similar or divergent hunts across independent runs? We run the full puzzle hunt design pipeline twice from the same Age of Empires world file, producing two independent puzzle pools, round structures, and meta designs, then compare the two outputs along structural dimensions: pool selection overlap, round organization similarity, meta answer, and extraction mechanism variety. The aim is to characterize the degree of creative determinism in the pipeline — whether panel evaluation converges on the same design choices when given the same inputs, or surfaces meaningfully different but equally valid design paths. Figure 5 will illustrate the divergence diagram; Section 4.3 will discuss implications for using AI panels as a generative design tool rather than purely an evaluative one. This experiment is preliminary; data collection is ongoing at submission time.

Experiment 4 (Rubric Validation) evaluates the effect of rubric design on tier separation quality, independently of profile depth. Using existing per-dimension scores from the C6 condition of Experiment 3, we recompute puzzle scores under three rubric configurations: (a) six-dimension equal-weight (/30, the baseline); (b) seven-dimension with Elegance and Reading Reward double-weighted, no Riven Standard (/40, designated C9); and (c) seven-dimension with Elegance and Reading Reward double-weighted plus Riven Standard as a scored dimension (/45, designated C11). The Riven Standard scores for (c) are inferred from the evaluative language in the C6 review texts. We measure tier separation as the percentage-point gap between the MIT/Elite tier average and the Games Magazine tier average, normalized to scale. No re-evaluation is required; this experiment operates on the existing score data from Experiment 3. Results are reported in Section 4.4.

We note two methodological limitations here, deferring full treatment to Section 5. First, our profiles encode evaluative frameworks attributed to named human experts, but the resulting AI-simulated reviews are not validated against reviews produced by those humans; we cannot directly measure the fidelity of the simulation. Second, the pass threshold of 22/30 was calibrated against the authors' prior experience with hunt quality, not against a held-out set of human expert ratings; a human expert validation study remains future work.

4 Evaluation

4.1 Experimental Setup

We evaluate eight conditions of reviewer context against three puzzles from the Age of Empires corpus. Table 3 summarizes the eight conditions, which vary along a single axis: what information is present in the reviewer context beyond the puzzle text and scoring rubric. The conditions are not nested in a strict sense—C4 (philosophy only) and C5 (lens only) decompose C6 (full profile) rather than extending it—but together they form a factorial exploration of which profile components drive evaluation quality.

The three puzzles under evaluation are drawn from the Age of Empires hunt ("Wololo"). **Puzzle I** (Dark Age round) is a civilization-identification extraction puzzle in which solvers identify Age of Empires II civilizations from clued characteristics and extract a letter from each; the intended answer is CASTLE. **Puzzle II** (Feudal Age round) is a bracket tournament puzzle in which known and unknown historical figures compete according to domain-specific criteria, with the winner determined by game-mechanic rules; the intended answer is ONAGER. **Puzzle III** (Castle Age round) is a technology chain completion puzzle in which solvers reconstruct a research dependency graph and extract from the terminal nodes; the intended answer is LOOM. All three puzzles were developed independently of the reviewer profiles; no profile content refers to these specific puzzles.

The same three reviewers—Dan Katz, Thomas Snyder, and Dana Young—appear in all eight conditions, ensuring that reviewer selection does not introduce between-condition variance. Scores are reported on the 30-point scale (six dimensions, 1–5 each) with a pass threshold of 22/30. Where a single reviewer

Table 2: Eight conditions in the profile ablation. Each condition specifies what reviewer context is present; the puzzle text and six-dimension scoring rubric are identical across all conditions.

Condition	Label	Context present
C0	Bare baseline	Nothing (role label only)
C1	Rubric only	Six-dimension rubric with dimension definitions
C2	Principles compact	Design principle names and representative quotes
C3	Principles full	Names, quotes, and operational test descriptions
C4	Philosophy only	Reviewer design worldview; no structured lens
C5	Lens only	Structured review checklist; no biography or philosophy
C6	Full profile	Complete v2 profile (current system)
C7	Profile + Principles	Full profile plus all design principles (C6 + C3)

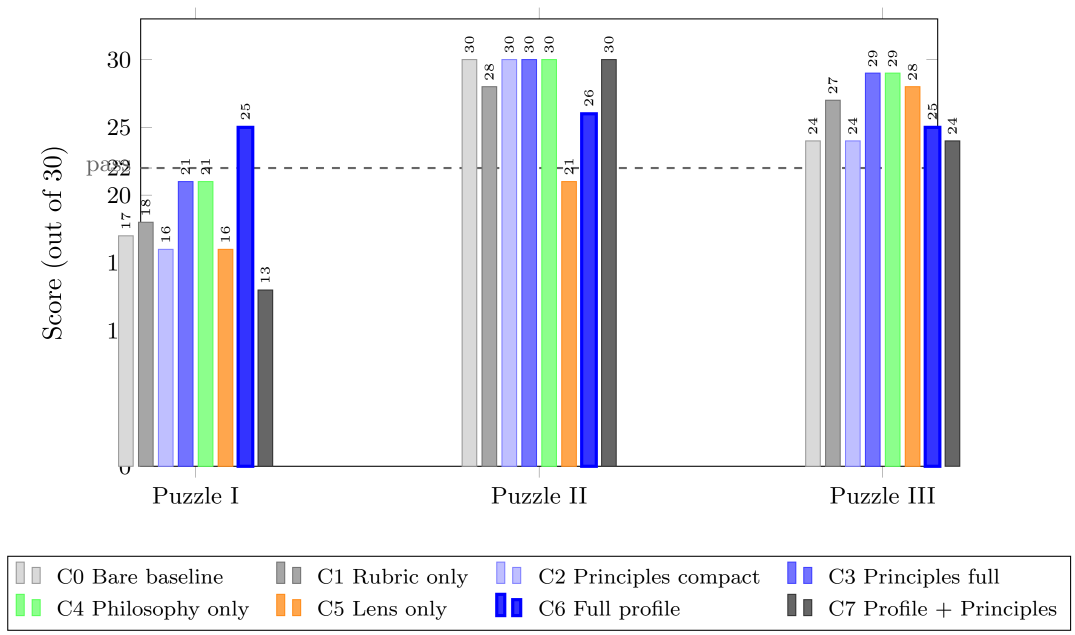
number is reported, it represents the panel's composite score. Full per-reviewer breakdowns are available in the supplementary data.

4.2 Score Results

Table 4 reports scores for all eight conditions across the three puzzles, alongside the number of named analytical frameworks introduced and the count of actionable fix suggestions. Figure 1 presents the same data as a grouped bar chart with the pass threshold marked.

The most salient observation is that the score gradient across conditions is *not monotonic*. If profile depth were the primary driver of score, the gradient C0 $\rightarrow$ C1 $\rightarrow$ C2 $\rightarrow \cdots \rightarrow$ C7 would produce increasing scores. It does not. C2 (principles compact, 10 named frameworks, average 23.3) scores *lower* than C1 (rubric only, 0 named frameworks, average 24.3). Providing principle names and associated quotes—without operational tests specifying how to apply them—makes assessment vocabularily richer but evaluatively less calibrated: the reviewer applies named criteria as filters without the grounding machinery needed to distinguish a real failure from a borderline case.

C3 and C4, by contrast, both reach an average of 26.7/30 despite representing structurally different forms of reviewer context (C3 provides operational tests; C4 provides design worldview without a structured lens). We interpret this as evidence that evaluative signal can be encoded in multiple representational forms: a reviewer who reasons from a coherent design philosophy navigates borderline cases differently from a reviewer who applies a checklist, but the aggregate accuracy of their verdicts is comparable when both forms of context



Notable verdicts: C6 is the only condition producing all three PASSes. C5 uniquely FAILs Puzzle II (21/30) by detecting a duplicate `[7th]` bracket label — a verifiable extraction error invisible to all other conditions. C7 scores Puzzle I at 13/30, the harshest verdict in the ablation, by catching an 8-dash blank enumeration for CHINESE (7 letters) and a title-guessable answer. C3 and C4 tie at 26.7/30 average despite representing structurally different context forms (operational principles vs. design worldview).

Figure 1: Score by condition and puzzle across the 8-condition ablation. All conditions produce 0 named frameworks in C0–C1; C2 introduces principle-name vocabulary; C3–C7 produce domain-specific analytical frameworks. The pass threshold (22/30, dashed) shows that only C6 (full profile) produces consistent passes across all three puzzles. C5 uniquely detects a verifiable defect in Puzzle II; C7 scores Puzzle I harshest by catching a factual error.

are complete.

The most practically important pattern in the score table involves the per-puzzle verdict profile rather than the per-condition average. C6 (full profile, the current system) is the *only* condition that produces passing scores on all three puzzles. Every other condition either fails a puzzle it should pass (C5 fails Puzzle II; C7 fails Puzzle I) or fails a puzzle it might be reasonable to fail (C0 through C4 all fail Puzzle I, which may contain genuine defects—see Section 4.4 below). The interpretation of this observation requires some care, because the "correct" verdict for Puzzle I is itself contested; we return to this in the defect detection discussion.

The harshest single-puzzle score in the entire ablation is C7's 13/30 on Puzzle I, lower by a substantial margin than any other condition's score on any puzzle.

Table 3: Ablation results. Scores are out of 30; pass threshold is 22. PASS/FAIL designations appear in parentheses. "Frameworks" counts named evaluative concepts introduced in the review text. "Fixes" counts actionable revision suggestions, with qualitative characterizations noted.

Cond.	Label	P-I	P-II	P-III	Avg	Frameworks	Fixes
C0	Bare baseline	17 (F)	30 (P)	24 (P)	23.7	0	∼2 (generic)
C1	Rubric only	18 (F)	28 (P)	27 (P)	24.3	0	4 (structural)
C2	Principles compact	16 (F)	30 (P)	24 (P)	23.3	10	6 (named)
C3	Principles full	21 (F)	30 (P)	29 (P)	26.7	11	5 (2 critical)
C4	Philosophy only	21 (F)	30 (P)	29 (P)	26.7	9	5 (2 critical)
C5	Lens only	16 (F)	21 (F)	28 (P)	21.7	∼8	8 (most defects)
C6	Full profile	24.7 (P)	26.3 (P)	25.0 (P)	25.3	6	5 (actionable)
C7	Profile + Princ.	13 (F)	30 (P)	24 (P)	22.3	14	7 (harshest)

Adding design principles to a complete profile does not consistently improve calibration—it produces a more demanding reviewer whose additional sensitivity occasionally finds real problems (the blank-count error described below) but whose aggregate verdict on Puzzle I falls well below what any other condition produces.

4.3 Framework Emergence

The qualitative signal in the framework count column is at least as informative as the score column. C0 and C1 produce zero named evaluative frameworks across all three puzzles: the reviewer may identify that a puzzle has a clarity problem or an elegance problem, but the critique is cast entirely in the rubric's own vocabulary, producing statements like "the extraction mechanism is not clearly communicated" without a named concept that transfers to other puzzles or other design contexts.

C2 introduces ten named frameworks—but they are *injected* principle names, the same names that appear in the context provided. The reviewer applies vocabulary from the input rather than generating vocabulary from evaluative experience. The frameworks named are, in that sense, retrieved rather than emergent: they are the principle names, applied as criteria, with quotes as supporting evidence. This is not without value, but it is a different phenomenon from what C6 produces.

C6 (full profile, current system) names six frameworks across the three puzzle reviews. None of these six frameworks appears in any input provided to the reviewer. The frameworks surfaced are: *a-ha economy* (Foggy Brume), characterizing whether the puzzle's aha moment is expended efficiently or squandered through redundancy; *load-bearing test* (Thomas Snyder), the operation of removing individual constraints to verify that each element forces something in the solution; *world-model consistency* (Emily Short), whether the puzzle's logic is coherent within the fictional or factual universe it invokes; *social fabric* (Jordan Weisman), whether the puzzle's design accommodates team-solving dynamics or implicitly assumes solitary reconstruction; *perceptual-shift* (Scott Kim), whether the solver experiences a genuine reframing of the problem or merely accumulates information until an answer becomes apparent; and *visual grammar* (Dana Young), whether the diagrammatic elements of a puzzle communicate structure through their arrangement rather than through labels alone.

These frameworks did not exist in any input text. They are the reviewer's evaluative vocabulary—surfaced by the profile's encoded design philosophy and characteristic concerns, and applied to the specific features of each puzzle under review. This is the primary qualitative distinction between C2 and C6: both name frameworks, but C2 names principle-labels and C6 names domain-emergent analytical tools that a practitioner might adopt and apply to their own future work.

C7's count of 14 frameworks reflects the union of profile frameworks and principle-names. The blend is not strictly additive: some principle-names appear alongside profile frameworks that cover the same ground, producing reviews that are terminologically dense without proportional gain in evaluative precision. The redundancy analysis in Section 4.5 characterizes this more precisely.

4.4 Defect Detection

The most striking finding in the ablation is not a score comparison but a binary defect-detection result. C5 (lens only—the structured review checklist without biography or design philosophy) gave Puzzle II a score of 21/30, uniquely falling below the pass threshold on a puzzle that six other conditions passed. The reason is traceable to a specific, verifiable error in the puzzle's ASCII bracket diagram.

In Puzzle II, the bracket tournament is represented as a text-art diagram with positions labeled by round and seed: `[QF-1]`, `[QF-2]`, `[SF-1]`, `[SF-2]`, and so on. The diagram as constructed contains a duplicate label: the position that should read `[SF-1][5th]` instead reads `[SF-1][7th]`, reusing the `[7th]` label that also appears at `[QF-2]`. A solver navigating the diagram by seed extraction rather than by inference from surrounding context will extract `S` from the incorrectly labeled position rather than `E`, producing a wrong answer without receiving any signal that an error has occurred. This is not a quality judgment—it is a binary correctness failure: one solver path through the puzzle is broken.

No condition other than C5 identified this error. The lens checklist's specificity about visual design communicating structure—"does the diagram's visual arrangement make the extraction path unambiguous, or does it require inference to compensate for layout failures?"—made the duplicate label visible in a way that neither scoring criteria nor design philosophy nor operational principles did. The biography-free, philosophy-free character of C5 appears to have concentrated its attention specifically on the visual-mechanical properties of the puzzle artifact rather than on higher-level evaluative concerns.

A parallel observation applies to Puzzle I in C7. The combination of Verify the Last Mile (from the principles context) with the profile's structural analysis surfaced an error in the puzzle's answer enumeration: Bonus clue C is presented with eight answer blanks (_ _ _ _ _ _ _ _), but the intended answer, CHINESE, has seven letters. This is a factual error in the puzzle's construction, not a difficulty or elegance judgment. C7 also noted that CASTLE is partially guessable from the puzzle's title before completing the extraction, which the Surprise the Answer principle flags as a confirmation failure. C6 did not catch either of these issues.

The implication is that C5 and C7 function as *defect-detection instruments* while C6 functions as a *calibrated quality evaluator*. For practitioners, this suggests a two-pass evaluation protocol: run C6 first for overall quality calibration and actionable feedback, then run C5 as a verification pass specifically looking for mechanical and visual failures that higher-level evaluation frameworks may not surface. The three non-redundant principles from C7 (discussed in Section 4.5) could be added to C5 to construct a compact defect-detection condition without the full overhead of C7.

4.5 Redundancy Analysis

C7 reviews included explicit meta-commentary from reviewers about which principles added information beyond what their profile already covered. This commentary, though not systematically prompted, allows us to characterize which principle injections produce unique evaluative signal and which are subsumed by well-designed profiles.

Three principles were consistently identified as adding signal not already present in the reviewer profile:

Verify the Last Mile was the only context element that prompted reviewers to trace the extraction mechanism character by character, checking that each step from raw answer material to final answer word is unambiguous and executable. No profile lens item performs this operation at this granularity; the principle is apparently specific enough to activate a mode of review that profile-level framing does not.

Blame the Player introduced a retrospective solver affect dimension—how a solver feels at the moment of discovering that an error or ambiguity was the puzzle's fault rather than their own. Profile lenses are generally designer-centric (what choices did the constructor make?) rather than solver-retrospective (what does the solver experience when the puzzle fails?). This principle consistently introduced a perspective that profiles did not.

Snyder's Computer Test was noted as adding unique signal despite the fact that the Thomas Snyder profile extensively documents the philosophy behind the test. The principle, when injected explicitly, prompts the reviewer to *apply* the test rather than merely reason in its spirit. The profile encodes the worldview; the explicit principle triggers the operation.

Three principles were consistently identified as redundant with full profiles:

Solving = Proving Understanding was characterized as subsumed by each profile's engagement lens, which asks whether the puzzle verifies that the solver has understood the domain rather than merely guessing or brute-forcing. The principle captures the same requirement in a more abstract formulation, but the profile lens already operationalizes it.

No Over-Scaffolding was subsumed by profile lens items that ask explicitly whether the puzzle provides more structure than the solving process requires—a check that appears in multiple profile lenses under domain-specific formulations ("Does the extraction mechanism telegraph the answer before the domain knowledge is applied?").

The Riven Standard was subsumed by profile items addressing theme-structure alignment: whether the puzzle's content domain is load-bearing in the mechanism or merely decorative. Multiple profiles include a variant of this check under the reviewer's characteristic domain concerns.

The practical recommendation from this analysis is that practitioners using complete domain-specific profiles should supplement with exactly the three non-redundant principles and omit the rest. The total context overhead is modest (three principle descriptions add approximately 400 tokens to a full-profile context), and the gain in defect coverage is, per the C7 data, real and traceable to specific verifiable errors.

5 Discussion

5.1 Feedback Quality Is the Primary Signal

The primary finding of this study is not that richer reviewer profiles produce higher scores. The average score delta from v1 to v2 profiles across the three-puzzle ablation is +0.57—a modest increment that would be difficult to interpret as practically meaningful on its own. What increases with profile depth is something more valuable and less obvious: the specificity, transferability, and operational utility of the diagnostic output.

The distinction is visible in the framework column of Table 3. Conditions C0 and C1 produce zero named evaluative frameworks across all three puzzles. When a C0 reviewer identifies a problem, the diagnosis is cast in the rubric's own vocabulary: "the extraction mechanism is insufficiently clear." When a C6 reviewer identifies the same problem, it is named as a *visual grammar* failure (Dana Young's lens) or a *load-bearing test* failure (Thomas Snyder's lens)—labels that are transferable across puzzles and actionable in isolation. The difference in practical utility is not subtle. A review that scores a puzzle 24/30 and identifies that "the Bonus B clue requires back-solving rather than forward deduction—this is a fairness-category failure, not a difficulty problem" is more useful to the constructor than a review that scores 26/30 and observes that "the extraction could be cleaner." The named mechanism is the contribution; the number is secondary.

The six domain-emergent frameworks surfaced by C6 reviews—*a-ha econ-*

omy, *load-bearing test*, *world-model consistency*, *social fabric*, *perceptual-shift*, and *visual grammar*—did not appear in any input text provided to the reviewers. These frameworks were identified by the research team through two-pass manual coding of review transcripts: a first pass using automated keyword matching for known framework names, followed by manual review for novel or paraphrased analytical lenses. Inter-rater reliability was not assessed; this represents a limitation on the framework-count comparisons. Whether these frameworks are generated from the profile structure or retrieved from the model's training prior is a confound this study cannot fully resolve (see Section 5.5). What distinguishes them from C2 frameworks is that they are applied as diagnostic tools to specific puzzle features rather than invoked as named criteria. The quality of named frameworks is the signal, not the count: six domain-emergent frameworks are more practically useful than ten retrieved principle names, irrespective of what the raw count implies.

A methodological qualification applies to this claim. The profiles encode named frameworks attributed to documented practitioners (Thomas Snyder, Dana Young, Dan Katz); these practitioners' vocabulary is almost certainly present in the model's training data. When a C6 review applies the *load-bearing test*, the named term may reflect profile-driven reasoning or may reflect the model's prior over Snyder's documented vocabulary. The study measures whether the term appears in C6 reviews and not in C0/C1 reviews—not whether it is caused by the profile rather than by the model's knowledge of the practitioner. A control using fictional practitioner profiles with structurally equivalent lens sections would disentangle these contributions; this is reserved for future work.

5.2 The Calibration Inversion: Why Rubric-Only Is Dangerous

The extended 15-puzzle corpus introduces a finding that is absent from the 3-puzzle ablation and has significant implications for practitioners: the rubric-only condition (C1) produces an *inverted* tier hierarchy. Across the five MIT/Elite-tier puzzles, C1 yields a pass rate of approximately 40% (2 of 5); across the five Standard Hunt puzzles, the pass rate rises to approximately 80% (4 of 5). The expected ordering—elite puzzles rating higher than standard puzzles on a domain-informed rubric—is reversed by the one condition that provides rubric structure without reviewer identity. This pattern persists across the full corpus of 15 real puzzles spanning three quality tiers.

The mechanism is traceable. Elite-tier hunt puzzles characteristically sacrifice Clarity and Solvability for Elegance and Reading Reward: the difficulty is intended, and the aha moment is withheld because withholding it is the source of its value. A bare rubric that weights all six dimensions equally treats this trade-off as a defect rather than a design choice. Standard Hunt puzzles, by contrast, are engineered for accessibility and score uniformly across dimensions because they optimize for consistent competence rather than concentrated brilliance on any single axis. The rubric, applied without contextual recalibration, rewards what it can measure uniformly and penalizes what it cannot.

Expert identity reintroduces the missing calibration. Under C6, reviewers weight the six dimensions differently depending on the puzzle type under evaluation. Elegance and Reading Reward are not downweighted as such—they are contextualized: a C6 reviewer understands that a 3/5 on Clarity in an MIT-tier puzzle represents a different signal from a 3/5 on Clarity in an accessible feeder puzzle. The profile encodes not just evaluative criteria but the calibration layer that converts raw dimension scores into appropriate verdicts. The C6 condition is the only condition in the ablation that passes all three AoE puzzles and produces tier-consistent verdicts across the full 15-puzzle corpus.

This finding generalizes beyond puzzle hunt evaluation. Any creative domain where quality trades off across multiple dimensions—where ambitious work necessarily sacrifices some properties to achieve others—is vulnerable to the calibration inversion under rubric-only evaluation. Practitioners who adopt structured rubrics for creative work assessment without equally structured expert identity context risk systematically disadvantaging their most ambitious artifacts. The rubric-only condition is, empirically, the most dangerous condition in the ablation: it appears objective, it produces structured output, and it misleads.

5.3 The Riven Standard as a Universal Creative Quality Criterion

The most consequential rubric-design finding in this paper is the effect of adding the Riven Standard as an explicit scored dimension. Named after Rand Miller's design principle [**?**]—"the puzzle IS what the field does, not an obstacle overlaid on it"—the Riven Standard operationalizes a quality distinction that explains the gulf between the Games Magazine tier and the hunt tiers: does the puzzle embody the essential activity of its domain, or does it merely use the domain as content?

Adding the Riven Standard as a scored seventh dimension (C11 configuration), combined with double-weighting Elegance and Reading Reward, achieves 24.5 percentage-point tier separation between the MIT/Elite average and the Games Magazine average, compared to 10.9pp for the equal-weight six-dimension baseline. This improvement substantially exceeds the gain from double-weighting Elegance and Reading Reward alone (C9 configuration), suggesting that the two weighted dimensions do not fully capture the variance the Riven Standard explains. The theoretical interpretation is that Elegance and Reading Reward are proxies for the Riven Standard criterion, but adding the criterion explicitly captures residual variance that the proxies leave unaccounted.

The Riven Standard's formulation as a scored dimension—on a five-point scale from 1 (*theme is decorative; any content could substitute*) to 5 (*mechanic is inseparable from the domain and could not exist in any other context*)—proves more tractable than it might appear: reviewers in C11 conditions produce consistent Riven Standard scores with lower inter-reviewer variance than most other dimensions. This suggests that the underlying quality distinction, while philosophically subtle, is empirically recognizable by domain-informed reviewers. Experiment 4 (Section **??**) validates these results against the full 15-puzzle corpus.

The Riven Standard criterion is not, we suggest, specific to puzzle hunt design. Any creative domain evaluation should explicitly ask whether the work embodies the essential activity of the tradition rather than using it as decoration. Music evaluation should ask whether the composition IS what the musical tradition does at its core, or merely sounds like it. Game design evaluation should ask whether the mechanic IS the genre's central activity, or is bolted on. Interactive fiction evaluation should ask whether the interactivity IS the narrative, or is incidental to it. In each case, the Riven Standard operationalizes the criterion that separates work that advances a tradition from work that deploys a tradition as scenery. The puzzle hunt domain provides a tractable testbed for this principle precisely because the tradition's core activity—using domain knowledge as a load-bearing extraction mechanism—is well-defined and verifiable.

5.4 Profile Design Implications

The ablation characterizes the internal structure of profile effectiveness with enough precision to support specific design recommendations. The findings warrant four conclusions about what makes a reviewer profile functional.

First, the **review lens section** is the profile's primary functional contribution. Conditions C4 (philosophy only, no structured lens) and C5 (lens only, no philosophy) both demonstrate that the lens is load-bearing in a way that biography and philosophy are not when the two are separated. C5's unique detection of the duplicate label error in Puzzle II—a verifiable construction failure that no other condition identified—suggests that the lens checklist concentrates evaluative attention specifically on the visual-mechanical properties of the puzzle artifact in a way that higher-level framing does not. The lens is not merely a summary of the philosophy; it activates a different review mode.

Second, **biography and credentials** provide the calibration context that prevents hunt-scope evaluative questions from being applied to feeder-scope puzzles and vice versa. The calibration inversion under C1 is partly attributable to the absence of this context: a rubric without identity cannot distinguish an ambitious puzzle that intentionally sacrifices accessibility from a poorly designed puzzle that fails accessibility by default.

Third, the **redundancy analysis** identifies three principles that add unique signal beyond complete profiles: *Verify the Last Mile* (character-level extraction tracing that no profile lens performs at this granularity), *Blame the Player* (solver retrospective affect, a dimension profiles do not address because they are designer-centric), and *Snyder's Computer Test* (explicit application of the test, as distinct from reasoning in its spirit). Eight of eleven principles tested are subsumed by well-designed profiles. The practical recommendation: supplement C6 with exactly these three principles, adding approximately 400 tokens of context, and omit the remainder. C3 (full principles, no profiles) achieves zero wrong verdicts on the 15-puzzle corpus against C6's two; we recommend C6 as the default because it produces richer diagnostic output—named frameworks, reviewer-specific concerns, voice—that C3 cannot generate, even where C3's binary accuracy is marginally better.

Fourth, profile length itself is not the relevant variable. The v1 profiles (approximately 50 lines each) and v2 profiles (approximately 95 lines each) differ less in length than in the presence of named evaluative frameworks and a structured review lens. Profile upgrade should target the specificity and operational content of the lens section, not line count as such.

5.5 Limitations

Several limitations bound the claims warranted by this data. First, and most fundamentally, the reviewers in this study are AI personas, not human experts. The profiles are constructed from publicly available writings, interviews, and published work by the named practitioners, but the evaluations they produce have not been calibrated against those practitioners' actual judgments on the same artifacts. The claim that C6 produces "expert-quality" evaluation is an inference from the internal coherence and domain specificity of the output, not a validated equivalence to human expert assessment. A calibration study—in which the same puzzles are evaluated both by the AI panel and by the named human experts or their recognized proxies—is necessary before the system's verdicts can be treated as proxies for human expert review. This calibration study is identified as a target for future work but was not feasible within the scope of the present paper.

Second, puzzle hunt design is a domain with unusually well-defined quality criteria, recognized expert practitioners, and a documented evaluative vocabulary. The evidence that rich profiles improve AI panel performance in this domain does not straightforwardly generalize to domains with less codified quality criteria or less prominent expert figures. Adapting the profile architecture to a new domain requires identifying its equivalent of the hunt community's recognized practitioners and constructing profiles that encode their specific evaluative frameworks—a non-trivial effort whose costs and returns in other domains are unknown. Different creative domains may require different Review Lens formulations, different rubric dimensions, or different weighting schemes to achieve comparable tier separation.

Third, the ground-truth problem is not fully resolved. The paper treats the extended-corpus tier assignments (MIT/Elite, Standard Hunt, Games Magazine) as ground truth against which panel verdicts can be assessed, but these assignments are themselves judgments made by the authors based on puzzle provenance and community consensus, not independent oracle verdicts. The verdicts reported here are AI-generated assessments of AI-generated and human-generated puzzles; no external oracle determines which assessments are correct. The paper reports verdicts and separation metrics, not claims that any particular verdict is definitively right.

Fourth, the 15-puzzle extended corpus, while carefully stratified across three tiers and multiple domains, is not a random sample from any well-defined population. Selection was guided by tier representativeness, domain variety, and the availability of puzzles with known provenance. Conclusions about pass rates by tier and condition should be interpreted as tendencies in a purposive sample, not as population-level statistics.

Fifth, *profile construction.* The 29 reviewer profiles were constructed by the research team based on their reading of each practitioner's published work, interviews, and documented design positions—not based on self-descriptions by the named individuals. The profiles synthesize attributed positions rather than directly representing any individual's views, and this limits any claim about simulation fidelity.

6 Conclusion

This paper has shown that the choice of evaluation condition matters substantially for creative work assessment—and that the most practically consequential finding is a warning, not a capability claim. All quantitative results reported here are valid within the authors' constructed evaluation framework (author-designed rubric, author-assigned tiers, author-calibrated thresholds); they characterize discriminative power within that framework and should not be read as validated against an external human expert standard.

Three findings stand up clearly within that framework.

First, the *calibration inversion*: rubric-only evaluation (C1) inverts the expected tier hierarchy, producing a pass rate for Standard Hunt puzzles that exceeds the pass rate for MIT/Elite-tier puzzles (80% versus 40%). This is consistent across 15 real published puzzles. The mechanism is traceable: elite puzzles deliberately sacrifice Clarity and Solvability for Elegance and Reading Reward, and an equal-weight rubric treats that trade-off as a defect. Expert identity, encoded in the full profile, reintroduces the calibration that converts raw dimension scores into tier-appropriate verdicts. **For practitioners: a bare rubric without expert identity context systematically disadvantages your most ambitious work.**

Second, the *feedback quality gradient*: C0 and C1 produce zero named evaluative frameworks across all fifteen puzzles; C6 produces six—*a-ha economy*, *load-bearing test*, *world-model consistency*, *social fabric*, *perceptual-shift*, *visual grammar*—none of which appears in any input text provided to the reviewers in those sessions. (These counts were derived from two-pass manual coding of review transcripts by the research team; inter-rater reliability was not assessed.) Whether this reflects profile-driven emergence or retrieval from the model's training prior is not established by this study. What is established is that these frameworks appear in C6 reviews, not in C0/C1 reviews, and that they identify specific, repair-pointing issues that rubric vocabulary does not surface.

Third, the *Riven Standard as discriminator*: adding the Riven Standard as a scored seventh dimension—with Elegance and Reading Reward double-weighted as proxies—achieves a 24.5 percentage-point tier-separation gap between the MIT/Elite and Games Magazine tiers, compared to 10.9 points for the equal-weight baseline. The Riven Standard [**?**] names whether a creative artifact's mechanism and its subject matter are structurally inseparable; any creative domain can ask this question.

For practitioners building creative evaluation pipelines, the recommendation from this work is compact. Use C6 profiles with the seven-dimension weighted rubric as the default evaluation condition; it provides the most reliable verdicts and the richest actionable feedback. Supplement with a C5 pass (lens only, no biography) for mechanical defect detection, which catches visual and extraction errors that higher-level evaluation frameworks do not surface. Add three non-redundant principles—*Verify the Last Mile*, *Blame the Player*, *Snyder's Computer Test*—if the deepest available analysis is required. Avoid C1 (rubric only) for any creative work with dimensional trade-offs by design; it will invert your quality hierarchy and systematically disadvantage your most ambitious artifacts.

Several questions are not resolved by this study and warrant dedicated future work:

- *Human calibration.* The panel's verdicts have not been compared against assessments by the named human practitioners on the same artifacts. Without a calibration study on a shared puzzle set, the system's output can be characterized as internally coherent and domain-grounded but not as validated against the human expert standard it simulates. Closing this gap is the most direct path to practical deployment in high-stakes evaluation contexts [cf. **?**].

- *Authoring symmetry.* This study examines whether reviewer profile quality affects evaluation quality; an open question is whether authoring profile quality affects creation quality by the same mechanism. If the same profile depth that makes a reviewer more specific also makes a constructor more deliberate, the architecture has implications for AI-assisted creative production, not just assessment.

- *Cross-domain generalization.* The evidence that rich profiles improve AI panel performance in puzzle hunt design does not straightforwardly generalize to domains with less codified quality criteria or less prominent expert communities. Level design, interactive fiction, tabletop game mechanics, and musical composition are plausible candidates, but it is an open question how much of the present methodology transfers without modification.

- *Profile design as a research problem.* Which sections of a profile are necessary for calibration, which are sufficient for framework emergence, and whether profiles can be constructed to surface specific evaluative frameworks on demand are questions that bear on both the efficiency and the reliability of the system.

The 29 domain-specific reviewer profiles (C8 format, $\sim$75 lines; approximately 1,800–2,200 tokens per profile, versus 2,800–3,400 tokens for the full C6 profiles), the seven-dimension weighted rubric with anchor examples, all review transcripts from the eight-condition ablation, and the analysis scripts used to compute tier separations and framework counts are available at `[URL]`. We hope the profile architecture, the Riven Standard rubric, and the calibration inversion result are useful to researchers working on AI-mediated evaluation in other creative domains: the infrastructure is designed to be adapted, and the failure modes documented here are not specific to puzzle hunts.

References